

Smart Home With Google Assistant Using Node MCU8266

**A
REPORT
ON
MINI PROJECT(2EL31)
Bachelor of Technology
(Electronics Engineering)**

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CERTIFICATE

This is to certify that the Mini Project (2EL31) entitled “**Smart Home With Google Assistant Using Node MCU8266**” has been carried out by ***Bhautik Barad and Rohit Gupta*** in the 4th semester under my guidance in Electronics Engineering, Birla Vishvakarma Mahavidyalaya, Vallabh Vidyanagar, during the academic year 2022-23.

Guided by: **Dr Deepak L.Vala**

ACKNOWLEDGMENT

It is indeed pleasure and pride moment for us to undertake this Mini Project entitled “**Smart Home With Google Assistant Using Node MCU8266**”. We would like to thank Dr. D.LVala, and Prof T.D.Pawar, Head of Electronics department, Birla Vishvakarma Mahavidyalaya, Vallabh Vidyanagar, Anand to inspire us to undertake this project and also to support during this project. We express our sincere gratitude to whole devoted staff of mechanical department and all those who helped us directly or indirectly during this Project.

ABSTRACT

Home automation is becoming popular day by day due to numerous advantages. Currently home automation system is managed by Emails, texts or some other applications. My project provides an idea of using IOT for monitoring and controlling the home appliances via wireless medium or Wi-Fi.

So the system is accessible from anywhere and anytime. The voice command function will be given to control any appliances or devices at home. This will provide a better communication in automated home as compared to normal homes.

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Chapter 1:

Title:- Introduction:

1.1 Problem Summary :

Different types of home automation systems I see and also I know different controllers and etc. But the major issues which I see or I observe is that most of all home automation systems control by physical mode like switch, Text message, Email or etc. In some advance system one issues is that they all are control by any another app which available is play store or apk and it's dangerous for mobile security perpose

✧ Introduction:

In this trending world, Internet of Things is being given extreme importance. In that, Automation, leads to have less effort and much efficiency. By using IoT, we are successful in controlling the appliances in various areas, in which one of them is to control the home automation by using Node Microcontroller. We can also use other boards like raspberry pi, beagle bone etc.

In the present- day technology, the whole work is done through communication so the effective way of communication can be done through voice. Even though the technology is developing in our day to day life, there is no help coming into existence for the people who are physically not good on the basis of technology. As the speech enabled, home automation system deploys the use of voice to control the devices. It mainly targets the physically disabled and elderly persons. The home automation will not work if the speech recognition is poor. The speech given by the user will be given as input to the Microphone. Microphone recognizes the speech given by the person and sends it to the recognizing module.

It searches for the nearest word even if there are any disturbances in it. If the command (ON/OFF) is given, the action is done. Similarly, the line following robot functions with respect to the speech commands given to it. The line following robot moves forward and backward with the help of sensors and a motor driver board. Home is the place where one desires to be rest after a long tiring day. People come home exhausted after a long hard-working day. Some are way too tired that they find it hard to move once they land on their couch, sofa or bed. So, any small device/technology that would help them switch theirs lights on or off, or play their favourite music etc. on a go with their voice with the aid of their smart phones would make their home more comfortable. Moreover, it would be better if everything such as warming bath water and adjusting the room temperature were already done before

they reach their home just by giving a voice command. So, when people would arrive home, they would find the room temperature, the bath water adjusted to their suitable preferences, and they could relax right away and feel cosier and rather, feel more homely.

Human assistants like housekeepers were a way for millionaires to keep up their homes in the past. Even now when technology is handy enough only the well to do people of the society is blessed with their new smart home devices, as these devices costs are a bit high. However, not everyone is wealthy enough to be able to afford a human assistant, or some smart home kit. Hence, the need for finding an inexpensive and smart assistant for normal families keeps growing.

1.2 Aim/Objective of the project:

To build a smart home device which can be used to control the home appliances via internet.

1.3 Problem Specifications:

Wired or wireless systems, while in the house not very likely to face many problems. However, when controlling through the application locally or remotely, we can face problems such as delays in performance and that is frustrating. Delays could be due to poor internet connection and planning in the house.

If we are having battery operated devices and their usage is more than normal, then battery problems could be another issue for what we to deal with and also the voice-recognition of Google assistant is not up-to the mark if there is noisy Environment. Due to Interference of the nearby noise, the Google assistant commands may also not work properly.

Home Automation is still an expensive product, by Indian consumer standards, and there's a lot of scope of the prices to be whittled down by an order of magnitude at least in the next few years. This will happen as more companies manufacture the key components of Automation systems and economies of scale takes over.

1.3 literature review about the project:

When people think about home automation, most of them may imagine living in a smart home: One remote controller for every household appliance, cooking the rice automatically, starting air conditioner automatically, heating water for bath

automatically and shading the window automatically during night. To some extent home automation equals to smart home.

Krunal Bhatt(2022) have proposed:

“Home automation using voice via Google assistant. The spoken commands from google assistant sends message to micro-controller this micro-controller pass the message to relay which will switch On and Off the appliances.

Jaypal Vaghela (2022):

“Home Automation System Using Google Assistant” This project presents a design and prototype of Home Automation system that will use ESP8266 Wi-Fi module as a network provider in connecting with other appliances. Further we will connect the specific home to our database and it can be accessed from anywhere through a specific IP address or website.

Roshan Singh (2022):

“Efficient low cost supervisory system for Internet of Things enabled smart home.” .

Viral Science (2022):

Science which certainly requires a careful study before developing any suitable Home Automation System (HAS) that will accomplish its requirements. Nevertheless, the latest attempts at introducing Home Automation Systems in actual homes for all kinds of users are starting to be successful thanks to the continuous standardization process that is lowering the prices and making devices more useful and easier to use for the end user. Even so several important issues are always to be handled strictly before developing and installing a Home Automation System; factors like security, reliability, usefulness, robustness and price are critical to determine if the final product will accomplish the expected requirements.

1.4 Materials / Tools required:

Hardware Requirements:

- 1.NodeMCU – 32-bit ESP8266 development board with Wi-Fi SoC.
- 2.Soldering Kit
- 3.One 15W Bulb
- 4.One 9V DC Fan
- 5.Resistor (As per requirements)
- 6.BC547 Transistor
- 7.IN4007 Diode
- 8.DC socket

- 9. 5v Relay
- 10. LEDs
- 11. Switch
- 12. General Purpose PCB

Software requirements:

- 1. Easy EDA (PCB Designing):
- 2. Proteous (Simulation):
- 3. Mit app inventor (Application Designing):
- 4. Arduino IDE (Coding):
- 5. Sinric Pro (Interconnecting Website):
- 6. Google Assistant (Commanding):

Chapter 2:

Title: Design

2.1 Simulation Designing:

As we all know, and I also say before that My project simulate in Proteous (Version 8.11) and also Whole simulating circuit diagram designing in proteous software . But the major problem in simulating software proteous is that in proteous we code only arduino devices , Because of proteous is made for only Arduino family and arduino controllers . So, we include the library of NodeMCU ESP8266 in proteous and it is uploaded successfully .

Still the problem is that in proteous NodeMCU ESP8266 don't able to take code and run it. So, we have to search another way for simulation, Or change the software. So, from that problem I search many different types of simulating software for simulating my project but no any one software to code NodeMCU ESP8266 to take code and run it perfectly. So, No any other option that's why for simulating purpose I decide that I simulating my project in proteous but in place of NodeMCU ESP8266 I will use Arduino UNO and Bluetooth module HC-05 and made one application which can give command to Bluetooth module HC-05 and after that Bluetooth Module give command to circuit. Then Circuit will simulating. But the one problem is still their and that is , some time application will not able to connect with desktop and then circuit will not simulating.

Most experiments/tasks performed today normally require the physical presence of the operator in order to monitor and run the task. Imagine what it will be like to be able to perform other activities elsewhere while still carrying out and monitoring an important task where one is. This work designs and simulates a system using Proteus software that will act as mobile interface between a machine and the user or operator such that simple, specific and repeatable task could be performed by users without being on the same location with the machine.

➤ Software used:

The program used in controlling the system is written using C programming language. After the program has been written in C, it was compiled on Arduino IDE Compiler in order to generate the microcontroller compatible HEX files. Below Figure shows a portion of the HEX file generated by the Arduino IDE compiler; the compiler that provides a complete integrated tool suite for developing and debugging embedded applications running on Microchip PIC.

2.2 Theory of operation:

The hardware was designed on the Proteus software. The Proteus is a simulation package tool that permits real time debugging of codes and circuit on the system. This Proteus VSM tool even permits the real interface of the mobile station / modem with the computer system, permitting all necessary tests to be performed. The circuit was drawn on the Proteus and the HEX file generated from the code compilation on Arduino IDE compiler (Figure), is written into the microcontroller in the circuit (Figure 4). This microcontroller is embedded in the circuit designed on the Proteus simulation package.

The mobile station (phone) is interfaced with the computer via Bluetooth in order to communicate with the microcontroller embedded in the circuit and the port number on the UART port is changed to the COM port number on the phone. The message (instruction command) is sent from the remote phone to the phone, interfaced with the computer system and the running button on the simulation package is pressed to start the simulation. The microcontroller fetches the SMS from phone, decodes it, recognizes the phone number and switches on the relays that are responsible for controlling the appliances. The microcontroller also sends back the report to the remote phone through an SMS.

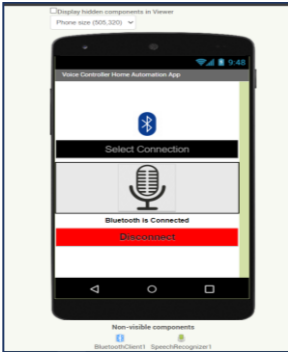


Figure 2: Layout of app



Figure 3: QR code of app

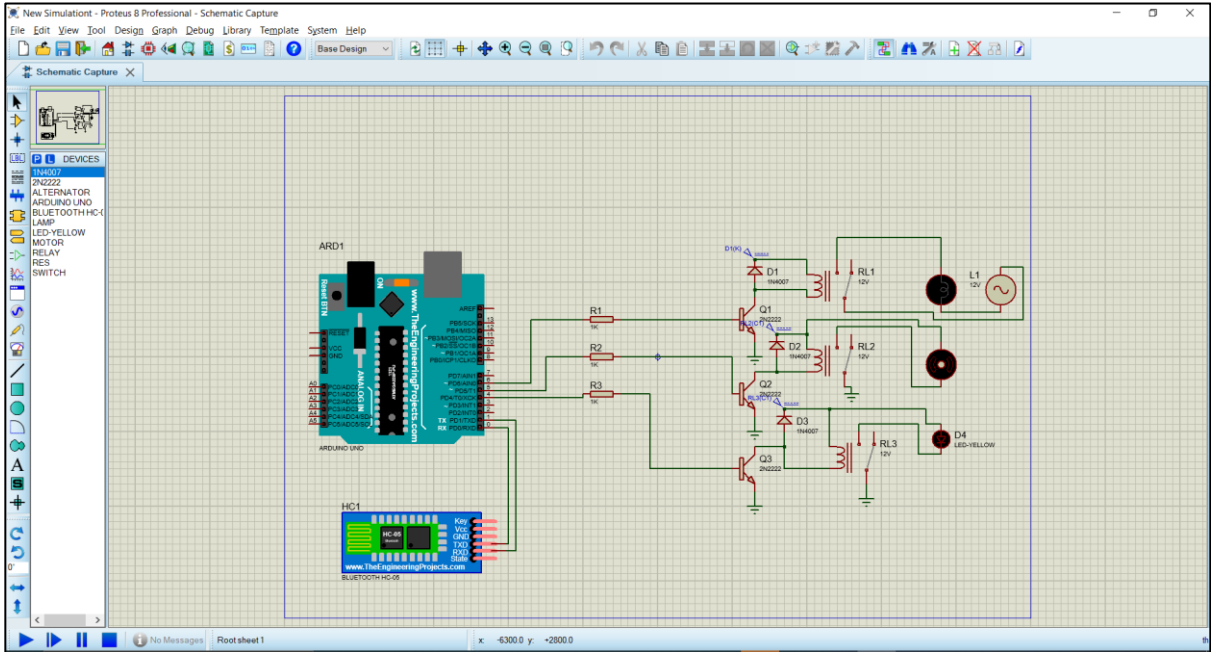


Figure 4: Before Simulation circuit diagram in proteous

RESULTS:

Figure 4 shows the circuit before the simulation is carried out. All three elements are not working, it is in its OFF state. In order to test the functionality of this system, there are two mobile stations. The remote one, that sends command and the one that is interfaced with the system, that receives the command. After the connection between the mobile station and the computer system has been established, the simulation button is triggered on and the simulation starts. Then the mobile station from a remote area is used to send the instructions (commands) to the mobile station connected with the system. The command is either to turn ON or OFF the home appliances. If it is to turn ON the home appliances.

After Simulating Results:

The operation of the system is confirmed (tested) by sending an SMS from the sending MS (Mobile Station) to trigger and to stop the operation of the appliance. The use of mobile communication system has the advantage of controlling the appliances remotely and it also makes life easy for the user.

For example, the user can get alerts anywhere through the GSM technology thus making the system location independent. Also, an SMS driven microcontroller for home automation is cost effective because you do not have to spend so much before you can control your appliances remotely. Once you have enough credit on your mobile system that can allow you to send the instructions to the mobile phone, connected to the system and the phone interfaced with the system also has credit that can allow it to send you back the reply. A lot of people have used similar methodology to control appliances, but this work does not only control the appliances, it also monitors the appliances. For example, the system monitors the level of water in the tank in order to know when to turn ON or OFF the pumping machine and monitors the home temperature, so that it will know when to switch on an Air Conditioner or not.

Here I attach some Photos of simulating Circuit.

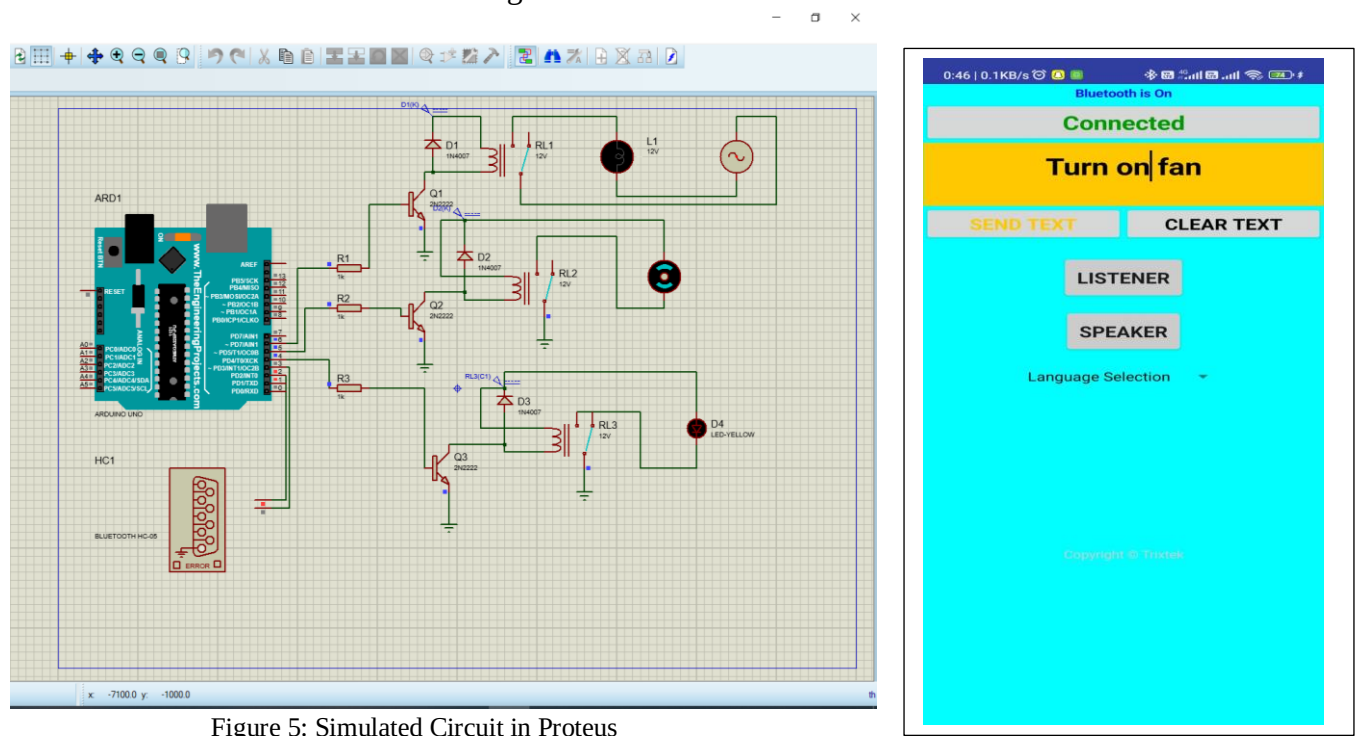


Figure 5: Simulated Circuit in Proteus

In figure 5 we see that the Fan is turning on and this command given by mobile application , In that application we see that we given command via text or voice . it's very easy to use. Now we see Next command that given by Mobile app.

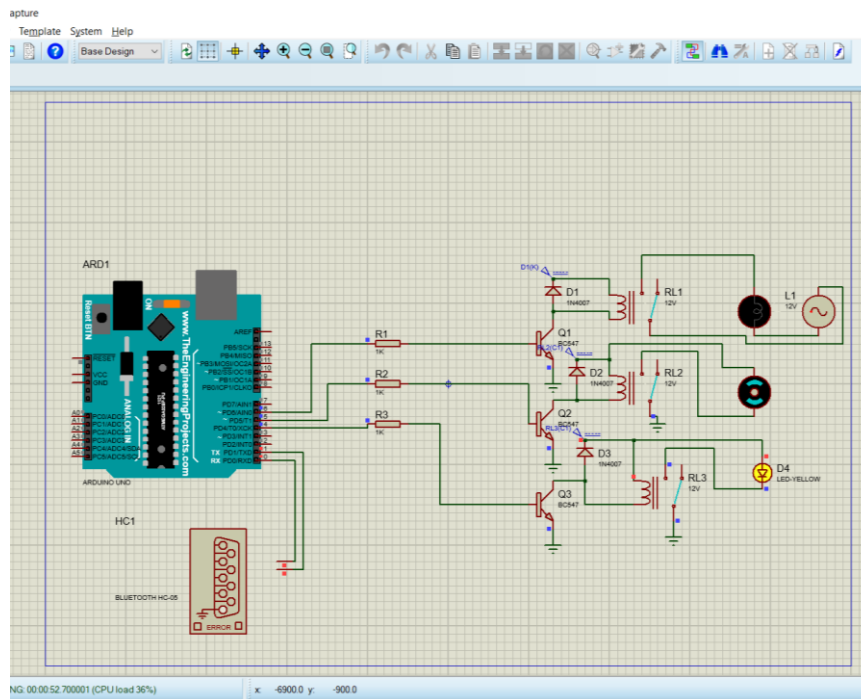
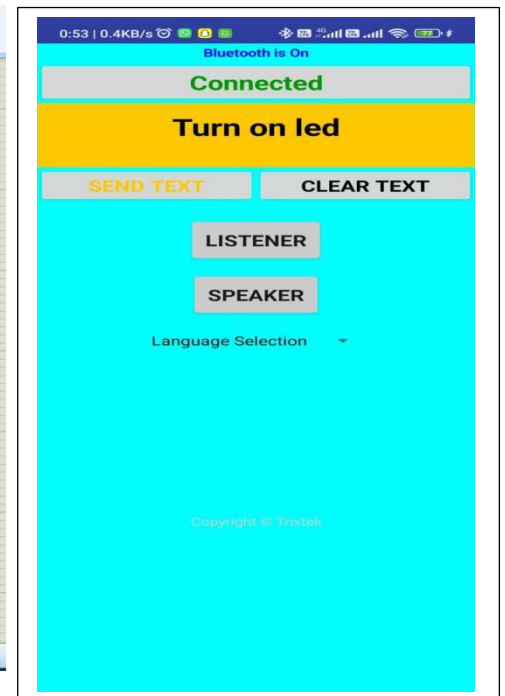


Figure 7: Simulated Circuit in Proteus



In figure 7 we see that as we given command to circuit that turn on LED and that command follows by circuit . So, circuit works perfectly. See next command,

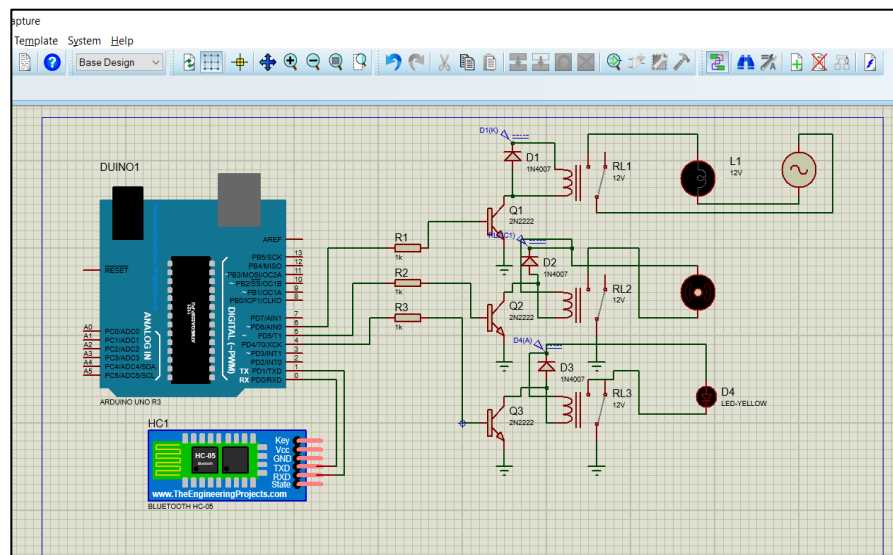


Figure 9: Simulated Circuit in Proteus

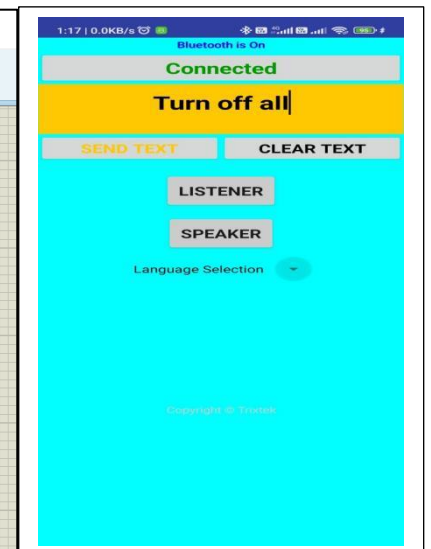


Figure 10: Commanding App

In figure 9 and 10 we clearly see and now also we understood that our circuit simulated properly and follow command properly. So, now that's all for software simulation and now we start Hardware simulation firstly on bread-board and then on PCB.

2.3 Methodology:

The methodology of this project design includes implementation of the proposed method. There are some basic steps involving in the Methodology of the product. The first major step is setting up the Sinric Pro Website. Sinric Pro is a website used to create virtual switches which will be turned ON or OFF depending on the commands given to the Google assistant and the second step is connecting the ESP8266 and the last step is connecting to Google assistant through Sinric Pro. Sinric Pro is a website used to create simple chain of conditional statements for like if else statements. By following these three steps, the implementation of the proposed system is going to be done.

2.4 Sinric Pro Website:

- First, create account at <https://sinric.pro/index.html#>

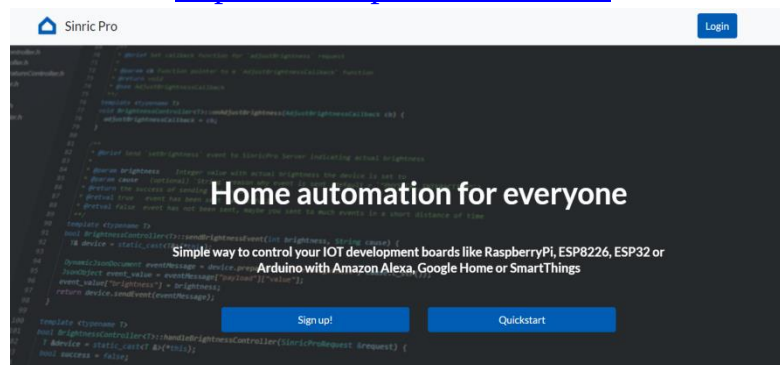


Figure 11: Sinric Pro Website Home Page

- Then go to the rooms and create your own room .

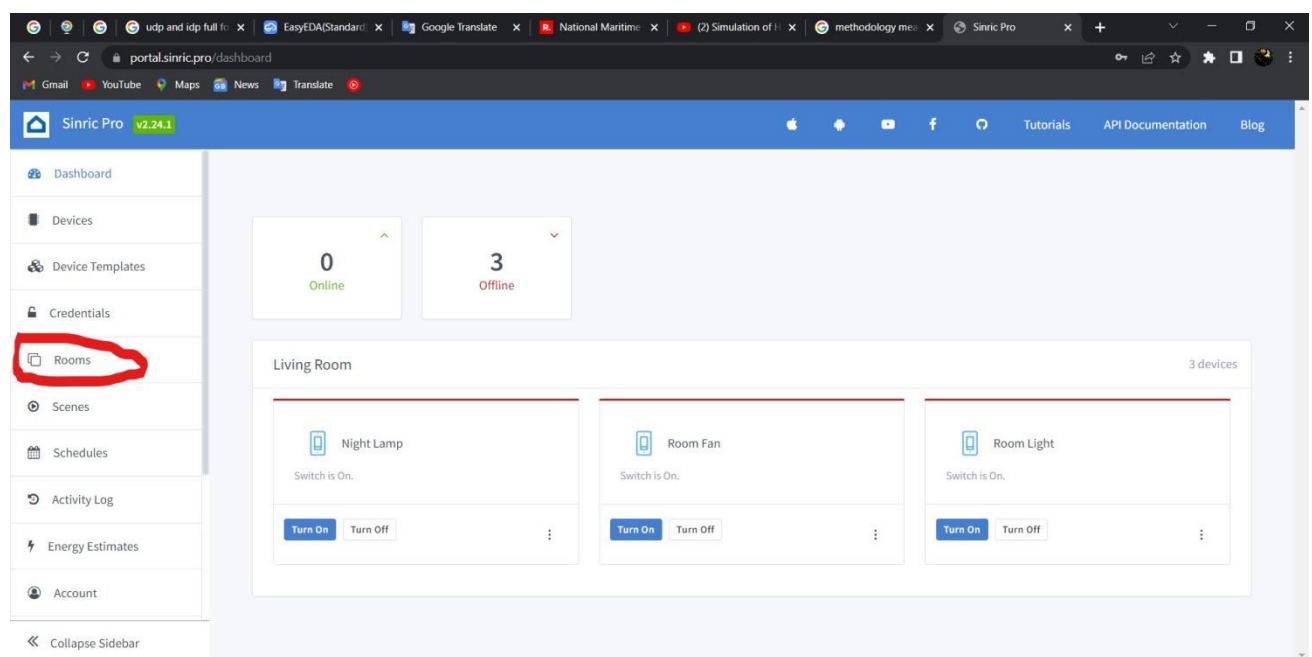


Figure 11: Sinric Pro website layout

- Then go to devices and as per your requirements create devices and in that their are many option like when device is connect or disconnect you give their notification and etc.

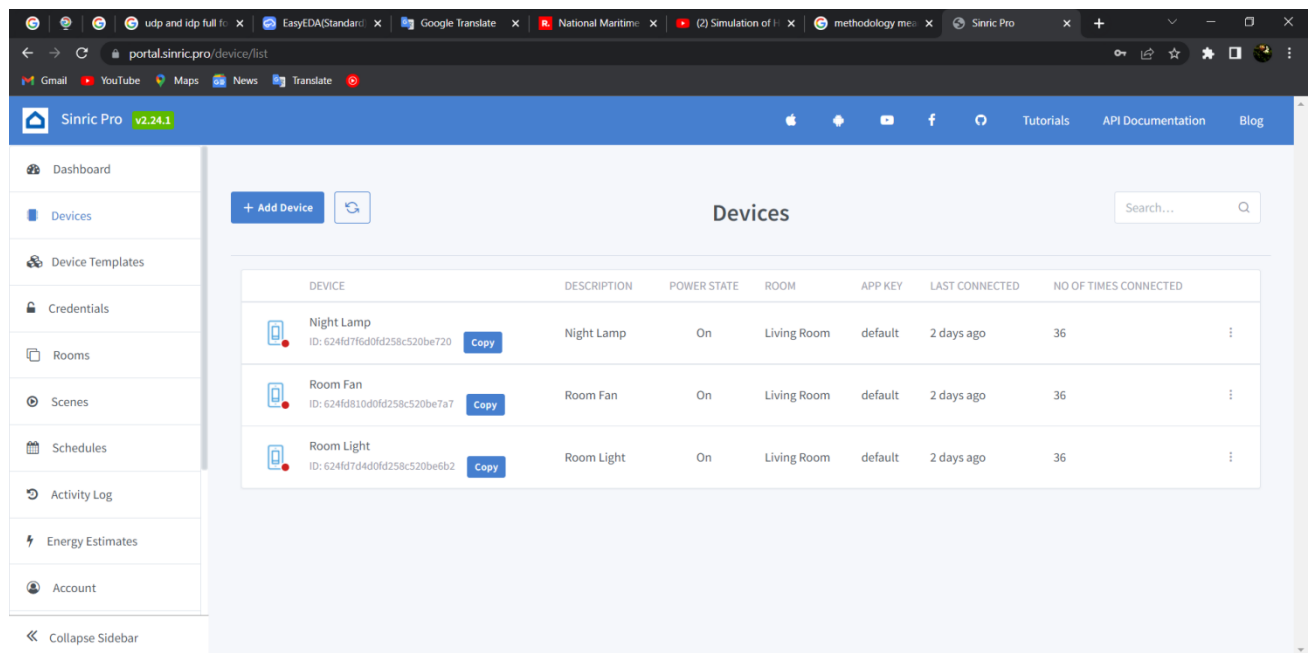


Figure 12: Lauout of device panel

- After that you go to dashboard and observe all think means notification and extra options which you choose , and one another think is that you also see your device is connect to you mobile or not . Device online means your device is connected and device is offline means your device is not connected.

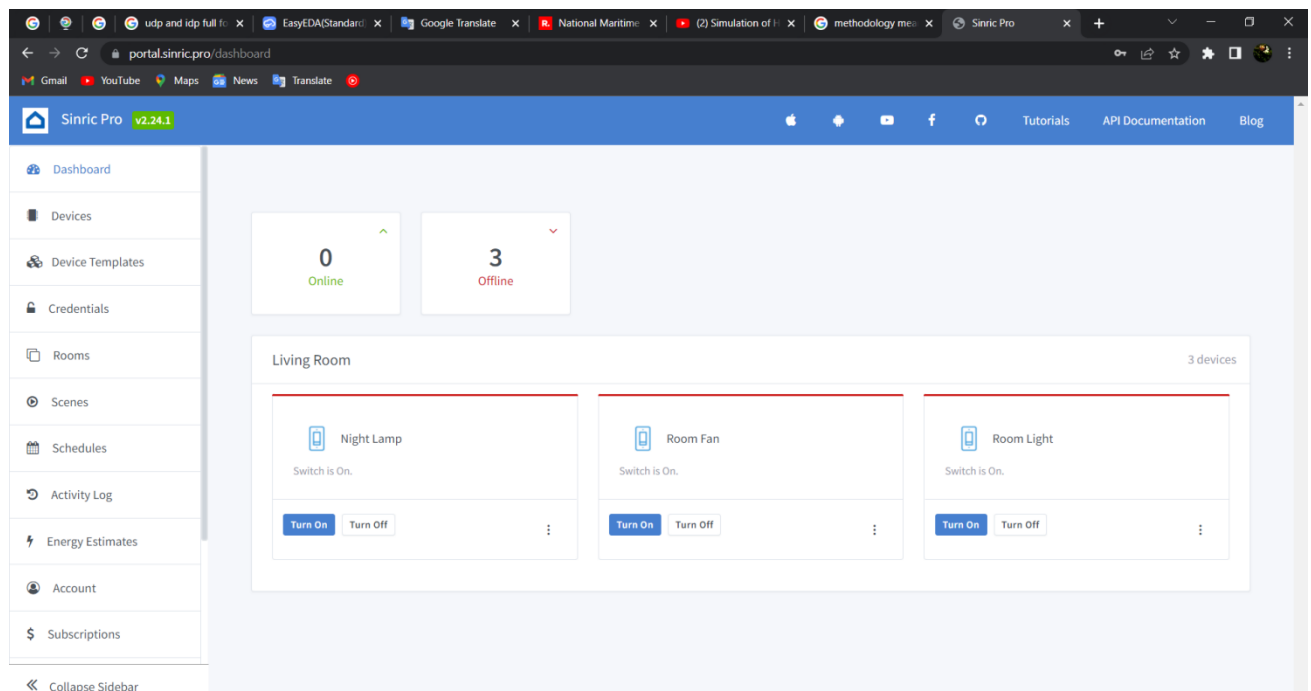


Figure 13: Dashboard of Sinric Pro Website

✧ Brief information about Sinric Pro:

With Sinric Pro, you can connect your IoT development board to Alexa or Google Assistant with a reliable skill - No more glitchy emulation. Sinric Pro supports all Amazon and Google IoT device types and offers Python, C++, NodeJS libraries with examples to get you up-and-running within minutes. For advanced users, you can create your own custom devices, set up rooms and routines, and monitor all your devices via REST API.

Sinric Pro enables developers to integrate IoT development boards (such as the RaspberryPi) with third-party applications or with Amazon Alexa and Google Home. The API can be used to retrieve device logs, find devices, update devices, and get account details. The Sinric Pro API features REST architecture, Tokens for authentication, and JSON responses.

Sinric Pro enables developers to integrate IoT development boards (such as the RaspberryPi) with third-party applications or with Amazon Alexa and Google Home. The API can be used to retrieve device logs, find devices, update devices, and get account details.

Connect Sinric Pro with Google Home App:

1) Tap on the “+” icon, then select Set up device.

2) Tap on Works with Google.

3) Search for Sinric Pro, then tap on Sinric Pro.

4) Enter the email id and password used for the Sinric account, Then tap on Sign in.

Now, we understand the connection first thing is Nodemcu esp8266 is connect to the WiFi and our mobile means Google assistant or Alexa is connect to Sinric Pro website . When we given command to Google assistant , it's send it to Sinpro website and then Sinric pro give command to Nodemcu esp8266 using wifi . So, this cycle is repeat in every command. This will be sensed by the Node microcontroller and it will turn ON or OFF the relay depending on the commands. All this will be done over the Internet. In this, the relay will act as a switch and the Home appliances connected to the relay will be turned on or off. The number of Home appliances connected depends upon the number of relays.

2.5 Block Diagram:

The block diagram of the Google assistant-controlled Home Automation is shown in the Figure below.



Figure 14: Block diagram of Google assistant controlled Home automation

2.6 Troubleshooting:

There are a number of places that the connection between our voice and the light can break down. If the light isn't changing when the user speaks, there are a few things should be checked. The light should be turned ON or OFF when the user toggles the switch on the Adafruit IO dashboard. If not, the ESP8266 is either not connecting to the server, not subscribing to the feed, or not checking for the correct string values.

Check the Serial Monitor output of the ESP8266 device to find out. If the Google Assistant doesn't hear the user properly, use the Google Allo app, users can see what the Assistant heard, or can directly type the phrase which the user want it to interpret. If the Google assistant doesn't respond with the correct phrase, then Google account and Sinric Pro account aren't connected. Make sure that the same Google accounts for the Google assistant and Sinric Pro is used. If the Arduino IDE dashboard doesn't update when the IFTTT applet triggers, then Arduino IDE account and Sinric Pro account aren't connected. Double check on Sinric Pro to make sure that both the accounts have been linked.

2.7 Interfacing:

The interfacing diagram of NodeMCU with Relay module is shown in Fig 3.18. The +5v Vin pin of the NodeMCU is given to the Voltage pin of relay module. The ground pin of Node Microcontroller is connected to ground pin of the Relay module. The NodeMCU consists of 8 data pins, clock, reset, enable, transmitter, receiver, flash etc., If the 4-Channel relay is used, then the data pins D0, D1, D2, D3 are connected to the 4 data pins of the Relay in which D0 is used to control 1st relay, D1 is used to control the 2nd relay, D3 is used to control the 3rd relay and D4 is used to control the 4th relay. The output of the relay consists of 3 pins in which two of them are given to the output like bulb, fan etc., and the one is of no connection.

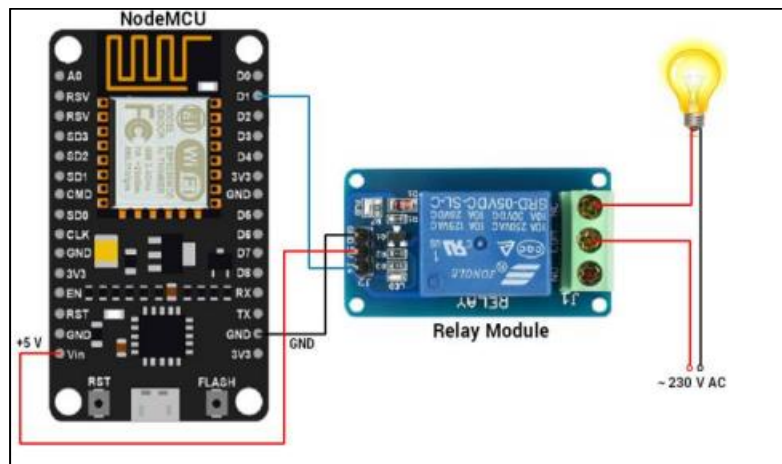


Figure 15: Interfacing Diagram of Node MCU (ESP8266) with Relay module.

Chapter 3:

3.1 Implementation:

The Implementation of the project design can be divided into two sections; Hardware and Software implementations. The hardware implementation consists of the development of the main controller, sensor networks and the smart home while the software implementation focuses on the programming of the Node microcontroller using Arduino IDE.

3.2 Components Description:

I. NodeMCU ESP8266:



Figure 16: Nodemcu ESP8266

NodeMCU is an open source IoT platform, includes firmware which runs on the ESP8266 Wi-Fi Module from Espressif Systems, and hardware which is based on the ESP-12 module. The term “NodeMCU” by default refers to the firmware rather than the development kits. NodeMCU firmware was developed so that AT commands can be replaced with Lua scripting making the life of developers easier. The ESP8266 is a low-cost Wi-Fi chip with full TCP/IP stack and microcontroller capability produced by Shanghai-based Chinese manufacturer, Espressif. Figure shows the NodeMCU (ESP8266) Development Board.

➤ FEATURES OF NODE MCU (ESP8266):

1. Open-source
2. Interactive
3. Programmable
4. Low cost
5. Simple
6. Smart
7. WI-FI enabled
8. USB-TTL included
9. Plug & Play

➤ SPECIFICATIONS OF NODE MCU (ESP8266):

1. Developer : ESP8266 Opensource Community
2. Type : Singleboard microcontroller
3. Operating system : XTOS
4. CPU : ESP8266
5. Memory : 128kBytes
6. Storage : 4MBytes
7. Power By : USB
8. Power Voltage : 3v ,5v(used with 3.3v Regulator inbuilt on Board using Pin VIN)
9. Code : Arduino Cpp
10. IDE Used : Arduino IDE

➤ ADVANTAGES OF NODE MCU (ESP8266):

1. Low energy consumption
2. Integrated support for WIFI network
3. Reduced size of the board
4. Low Cost

3.2 PIN DIAGRAM OF NODE MCU:

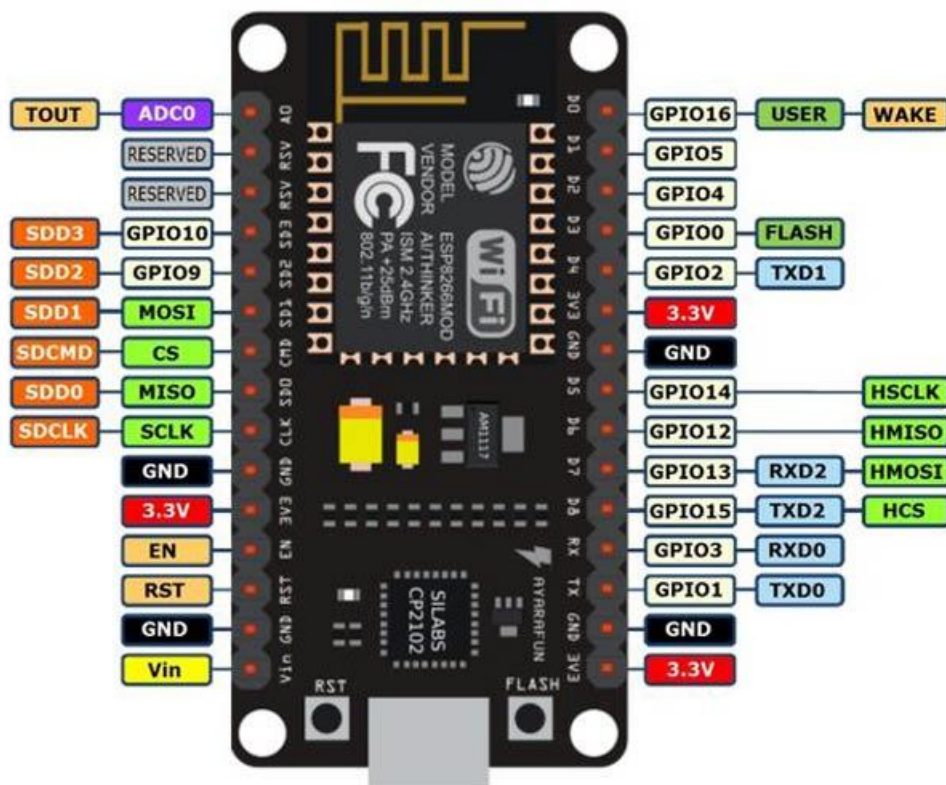


Figure 17: Nodemcu ESP8266 Pin Diagram

3.3 PROGRAMMING NODE MCU:

- Install the current upstream Arduino IDE at the 1.8 level or later. The current version is at the Arduino website.
- Start Arduino and open Preferences window.
- Enter https://arduino.esp8266.com/stable/package_esp8266com_index.json into Additional 1 Board Manager URLs field. You can add multiple URLs, separating them with commas.

After Completion,

1. Data Cable of your Mobile Phone. Used in To Connect ESP8266 MCU NODE with PC.
2. After Install Drivers if Needed.
3. Check Which Number Is Assigned To your Board.
4. Open Arduino IDE.
5. Open Boards Manager from Tools > esp8266 Modules platform And Select NodeMCU 1.0(ESP-12E Module) board from Tools.

- Upload Using: Serial
- CPU Frequency: 80Mhz
- Flash Size: 4M
- Upload Speed: 115200
- PORT: Select Assign Port Only.

II. RELAY MODULE:

A 4-Channel Relay interface board allows us to control various appliances, and other equipment's with large current. It can be controlled directly by Micro-controller (Arduino, Node MCU, Raspberry Pi, 8051, AVR, PIC, DSP, ARM, ARM, MSP430, TTL logic).

Specifications:

- 4-Channel Relay interface board, and each one needs 15-20mA Driver Current
- Both controlled by 12V and 5V input Voltage
- Equipped with high-current relay, AC250V 10A; DC30V 10A
- Standard interface that can be controlled directly by microcontroller.
- Opto-isolated inputs
- Indication LED's for Relay output status.

A relay is an electromagnetic switch. It is activated when a small current of some microampere is applied. Normally, a relay is used in a circuit as a type of switch, an

automatic switch. There are different types of relays and they operate at different voltages. When a circuit is built the voltage that will trigger it has to be considered. In this system the relay circuit is used to turn the appliances ON/OFF. The high/low signal is supplied from the NodeMCU microcontroller. When a low voltage is given to the relay of an appliance it is turned off and when a high voltage is given it is turned on. The relay circuit is to drive four appliances in the Home automation system. The number of appliances can be modified according to the user's requirements.



Figure 18: 4- Channel relay module (5v)

➤ Pin Description:

Pin Name	Description
"Vcc"	Power(5V DC)
"GND"	Gnd
"in1"	Singal pin, connected with Arduino and control Relay 1
"in2"	Singal pin, connected with Arduino and control Relay 2
"in3"	Singal pin, connected with Arduino and control Relay 3
"in4"	Singal pin, connected with Arduino and control Relay 4
"COM"	Common pin, which usually directly connect with the "Gnd" unless you want to change the TTL mode(default the HIGH level activate)
"NO"	Normally Open Connection
"NC"	Normally Closed Connection
"C"(middle pin)	Common Connection, Which connected with the power for the load.

Figure 19: Pin description

➤ ADVANTAGES:

1. Relays can switch AC and DC, transistors can only switch DC.
2. Relays can high voltages, transistors cannot.
3. Relays are a better choice for switching large currents(>5A).
4. Relays can switch many contacts at once.

3.4 SOFTWARE REQUIREMENTS:

1. Google assistant application
2. Sinric Pro
3. Arduino IDE

GOOGLE ASSISTANT:

The Google Assistant is an Artificial Intelligence based Virtual assistant software which allows its users to control all the apps in their device. It allows the users to control and command most of the apps in their devices using voice commands. This provides more convenience to the people as they only have to command the google assistant thorough voice command.



Figure 20: Google Assistant

Google Assistant is an artificial intelligence-powered virtual assistant developed by Google that is primarily available on mobile and smart home devices. Unlike the company's previous virtual assistant, Google Now, Google Assistant can engage in two-way conversations. Assistant initially debuted in May 2016 as part of Google's messaging app Allo, and its voice activated speaker Google Home. After a period of exclusivity on the Pixel and Pixel XL smartphones, it began to be deployed on other Android devices in February 2017, including third-party smartphones and Android Wear (now Wear OS), and was released as a standalone app on the iOS operating system in May 2017.

Alongside the announcement of a software development kit in April 2017, the Assistant has been, and is being, further extended to support a large variety of devices, including cars and third-party smart home appliances. The functionality of the Assistant can also be enhanced by third-party developers. Users primarily

interact with Google Assistant through natural voice, though keyboard input is also supported.

In the same nature and manner as Google Now, the Assistant is able to search the Internet, schedule events and alarms, adjust hardware settings on the user's device, and show information from the user's Google account. Google has also announced that the Assistant will be able to identify objects and gather visual information through the device's camera, and support purchasing products and sending money, as well as identifying songs.

➤ Sinric Pro Website:

With Sinric Pro, you can connect your IoT development board to Alexa or Google Assistant with a reliable skill - No more glitchy emulation. Sinric Pro supports all Amazon and Google IoT device types and offers Python, C++, NodeJS libraries with examples to get you up-and-running within minutes. For advanced users, you can create your own custom devices, set up rooms and routines, and monitor all your devices via REST API.

Sinric Pro enables developers to integrate IoT development boards (such as the RaspberryPi) with third-party applications or with Amazon Alexa and Google Home. The API can be used to retrieve device logs, find devices, update devices, and get account details. The Sinric Pro API features REST architecture, Tokens for authentication, and JSON responses.

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Figure 22: Sinric Pro website

- ARDUINO IDE :

The Arduino integrated development environment (IDE) is a crossplatform application (for Windows, macOS, Linux) that is written in the programming language Java. It is used to write and upload programs to Arduino compatible boards, but also, with the help of 3rd party cores, other vendor development boards. The source code for the IDE is released under the GNU General Public License, version 2.

The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub `main()` into an executable cyclic executive program with the GNU toolchain, also included with the IDE distribution. The Arduino IDE employs the program `avrdude` to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware.



Figure 21: Arduino IDE

Components list:

<u>Component Name:</u>	<u>Component Quantity:</u>
NodeMCU ESP8266	1
2) Soldering Kit	1
3) 5v Voltage Regulator	1
4) 330ohms resistor	6
5) 220ohms resistor	4
6) BC547 Transistor	4
7) IN4007 Diode	4
8) DC socket	1
9) 5v Relay	3
10) LED	6
11) Push Buttons	4

3.6 Circuit Diagram of Project:

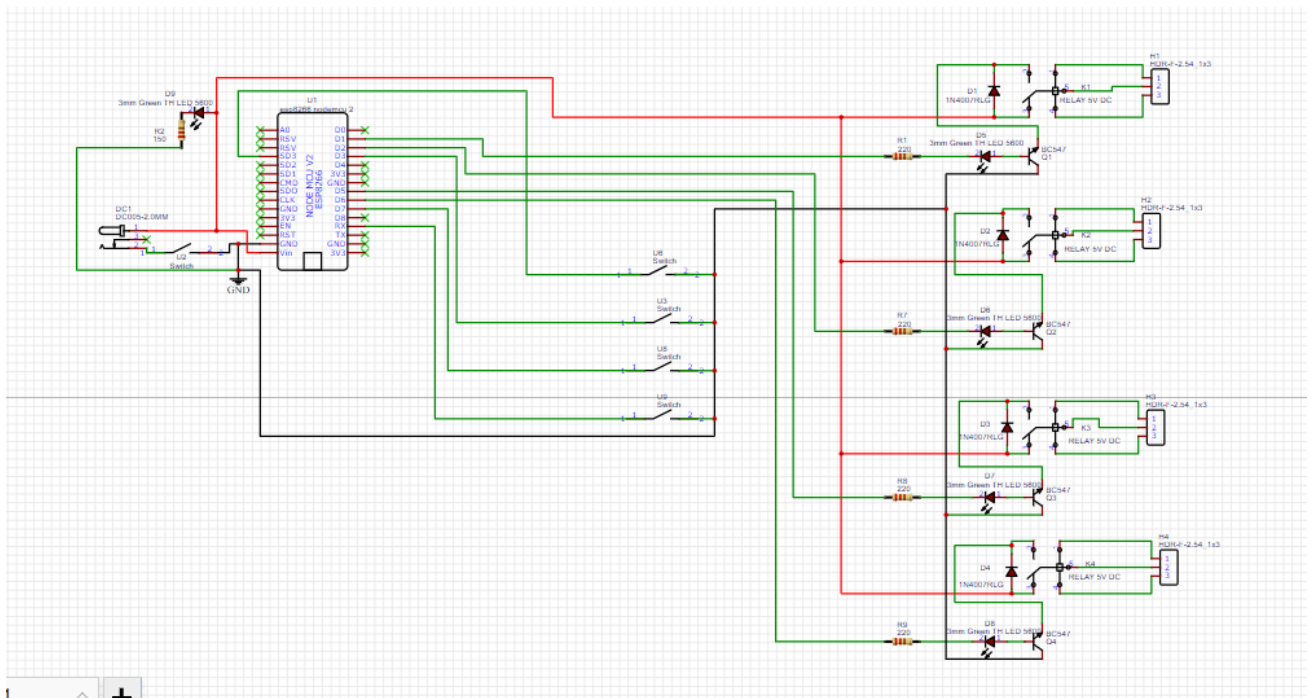


Figure 24: Circuit Diagram

3.7 Circuit Diagram of Relay Module:

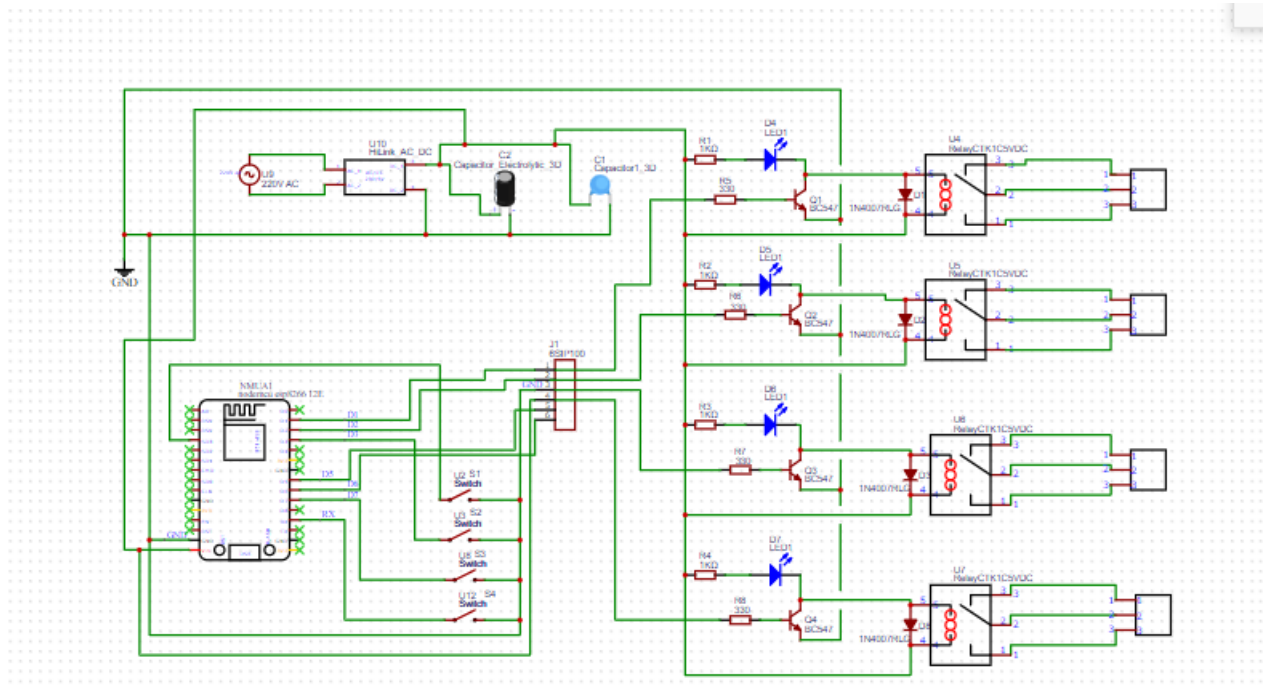


Figure 25: Relay module Circuit Diagram

3.8 Block Diagram:

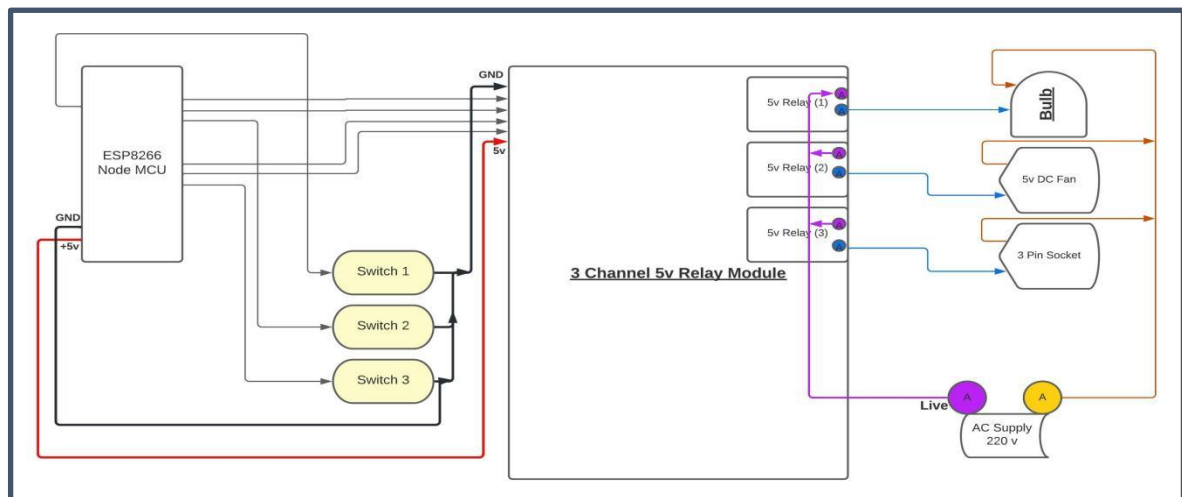


Figure 26: Block Diagram

3.6 PCB Designing:

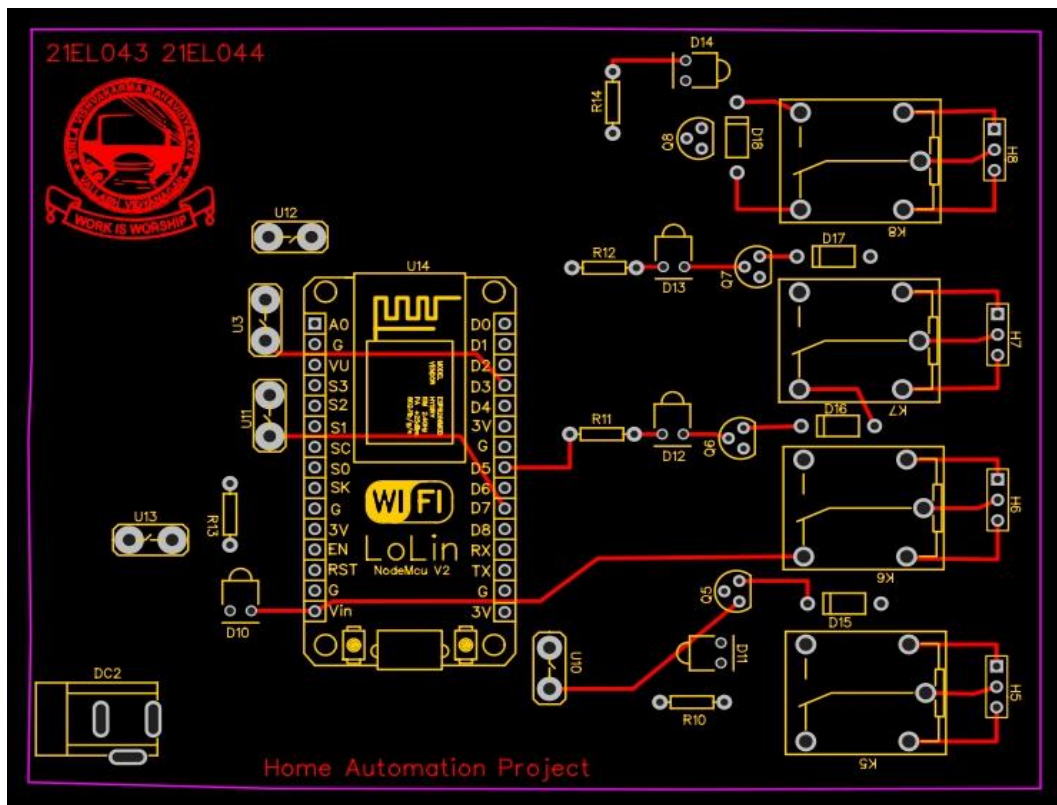


Figure : PCB layout

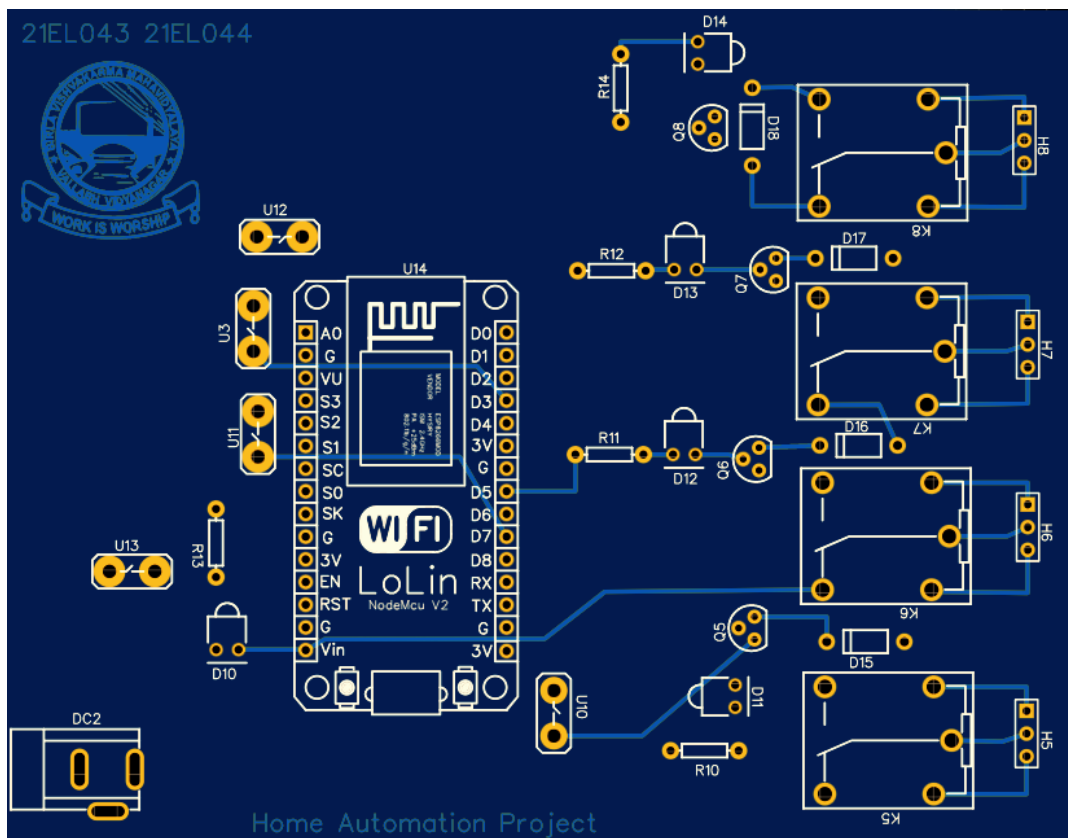


FIGURE:2D VIEW

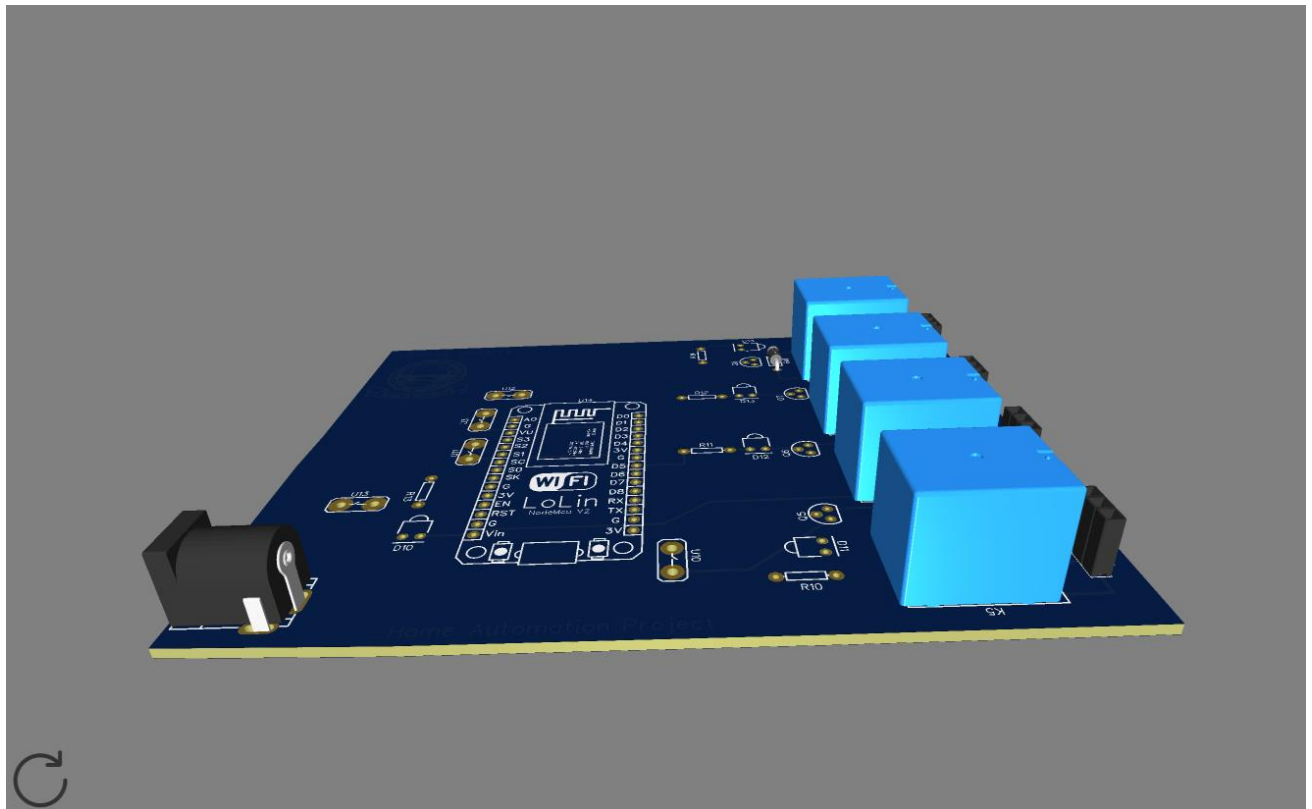
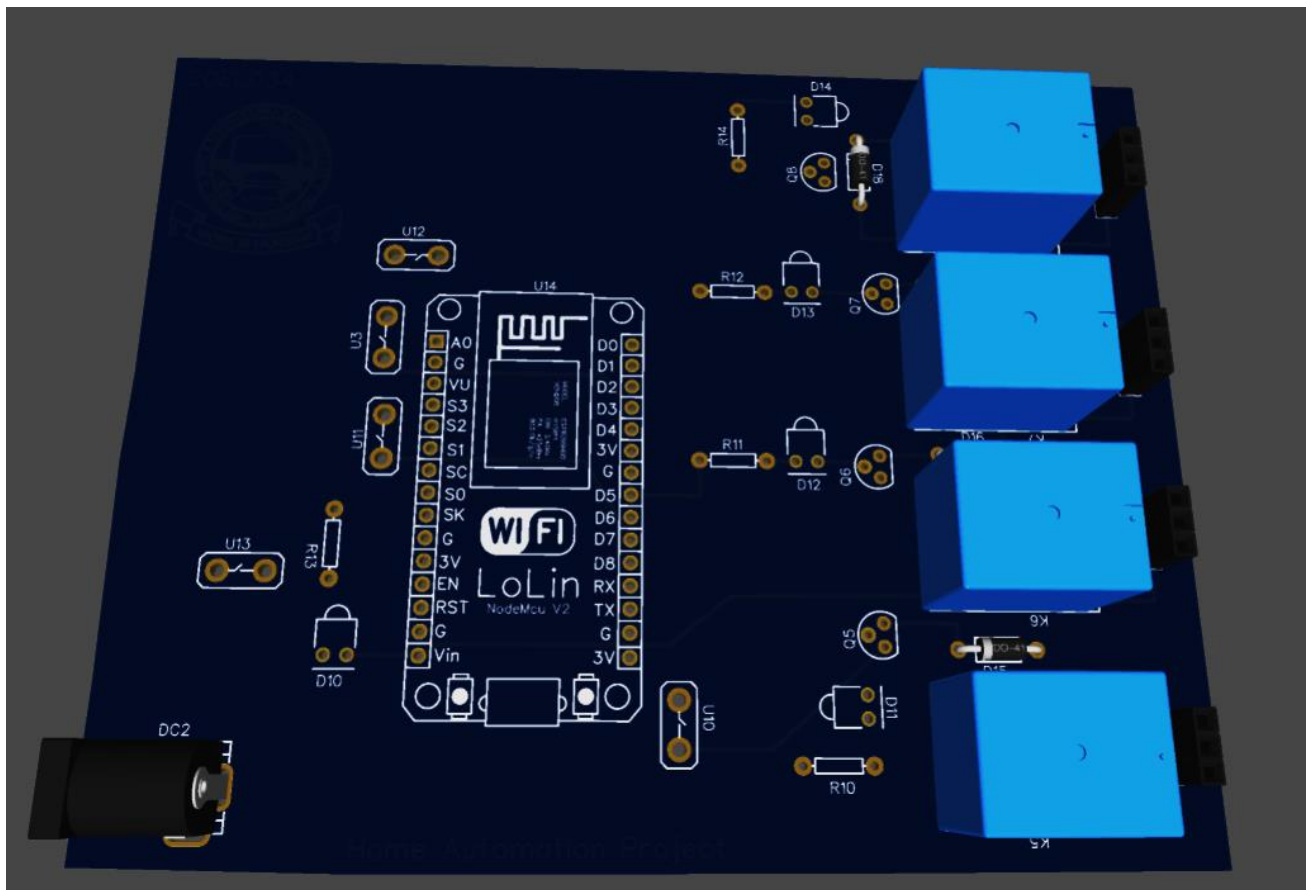


FIGURE: 3D VIEW

Applications:

- Home Automation will save time in daily recursive activities.
- Users can sit anywhere in the globe access our home away from home.
- Users will have the convenience of controlling devices through smart phone or tablet.
- Security, users can secure their home while they are away from home.
- There are devices which keep monitoring the home and notify them when any incident occurs.
- These devices can also run on Batteries. So need not Don't worry about even when electricity is turned off.

Advantages:

- Control home appliances from anywhere in the world.
- Easy to operate.
- Android compatible smartphone app.
- You will receive a email or sms for any activity.
- Saves electricity.

Limitations:

- Reliability
- System crashes due to any damage in the interconnection
- Costly

Future Work:

There are a variety of enhancements that could be made to this system to achieve greater accuracy in sensing and detection.

- a) There are a lot of other sensors that can be used to increase the security and control of the home like pressure sensor that can be put outside the home to detect that someone will enter the home.
- b) Changing the way of the automated notifications by using the GSM module to make this system more professional.
- c) A smart garage that can measure the length of the car and choose which block to put the car into it and it will navigate the car through the garage to make the parking easy for the homeowner in his garage.

Future scope for the home automation systems involves making homes even smarter. More energy can be conserved by ensuring occupation of the house before turning on devices and turning off lights if not necessary. The system can be integrated closely with home security solutions to allow greater control and safety for home owners. The next step would be to extend this system to automate a large scale environment, such as offices and factories.

Home Automation offers a global standard for interoperable products. Standardization enables smart homes that can control appliances, lighting, environment, energy management and security as well as the expandability to connect with other networks. Well, no system is ever perfect. It always has a scope for improvement. One just needs to put on a thinking cap and try and make the system more better.

Conclusion:

We propose a secure and low-cost system that can remotely control smart home appliances through an android application. The methodology we discussed in this paper is novel in achieving the target of controlling home appliances remotely using Wi-Fi technology. The proposed system not only monitors the sensor data but also takes necessary actions according to the circumstances, for example turning ON fans when the temperature crosses a threshold value. The approach we discussed in this paper also stores the sensor data in the database to analyse the various conditions of the user's environment. The application can be enhanced further by adding smart camera functionality to monitor the house and to get notified in case of any intrusion. Smart gardening can also be made part of a smart home where the sprinklers water the plants when they detect low moisture in the soil.

Reference:

- 1)<https://youtu.be/gpB4600keWA> (reference video)
- 2)<https://sinric.pro/index.html> (IoT Software)
- 3)<https://easyeda.com/> (PCB Design)
- 4)[Google-assistant-controlled-home-automation.pdf](#) (reference pdf)
- 5)<https://appinventor.mit.edu/> (App designer Software)
- 6)<https://youtu.be/EvAtsI59Gyc> (reference video)
- 7)<http://ai2.appinventor.mit.edu/b/5gtbg> (My App for simulation)
- 8) "Home Automation based on an Android and a web application using Raspberry Pi" by Dimitris Kehagias and Dorian Nini. "Design and implementation of Wi-Fi based smart home system" by Vinod Choudhary, Aniket Parab, Satyajit Bhapkar, Nitesh Jha and Ms. Medha Kulkarni.
- 9) Intelligent Smart Home Automation and security system using Arduino and Wi-Fi by J.Chandramohan,R.Nagarajan, K.Satheeshkumar,N.Ajithkumar,P.A.Gopinath and S.Ranjithkumar.